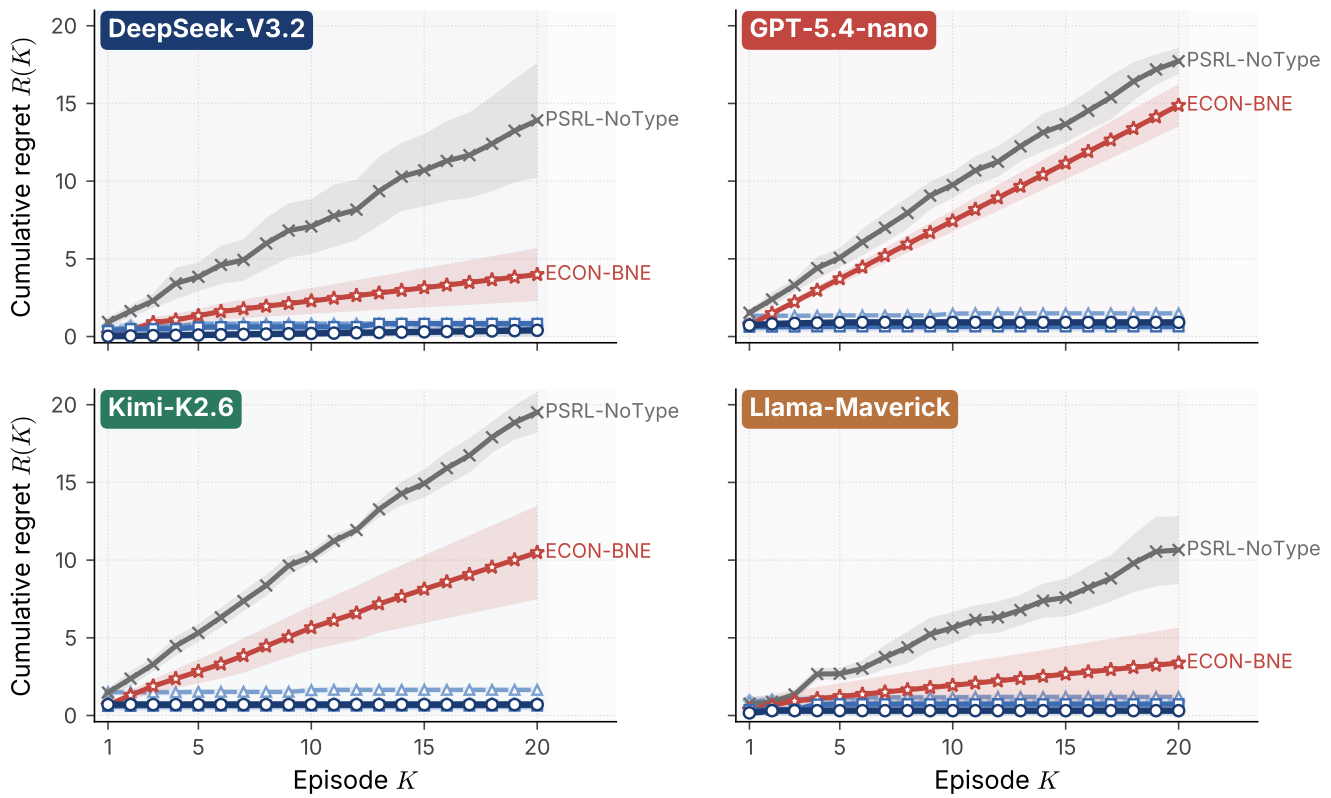


E5. Cumulative regret trajectories validate the predicted $\tilde{O}(\sqrt{K})$ rate and the rate separation from type-agnostic baselines on the LLM substrate.



—○— H-PSMG⁺ (ours) —□— H-PSMG (ours) —△— Joint-PSRL ···◇··· MAP-Greedy —×— PSRL-NoType —★— ECON-BNE

Ours (Bayesian, deep navy) plateau by $K \approx 5$, consistent with the $\tilde{O}(\sqrt{K})$ upper bound. PSRL-NoType (gray, type-agnostic) and ECON-BNE (red, LLM-coord) accumulate regret approximately linearly in K . Bands are ± 1 SEM over 5 seeds.